assignement

2023-03-30

Load dataset

```
## tibble [53,940 x 10] (S3: tbl_df/tbl/data.frame)
##  $ carat  : num [1:53940] 0.23 0.21 0.23 0.29 0.31 0.24 0.24 0.26 0.22 0.23 ...
##  $ cut    : Ord.factor w/ 5 levels "Fair"<"Good"<..: 5 4 2 4 2 3 3 3 1 3 ...
##  $ color  : Ord.factor w/ 7 levels "D"<"E"<"F"<"G"<..: 2 2 2 6 7 7 6 5 2 5 ...
##  $ clarity: Ord.factor w/ 8 levels "I1"<"SI2"<"SI1"<..: 2 3 5 4 2 6 7 3 4 5 ...
##  $ depth  : num [1:53940] 61.5 59.8 56.9 62.4 63.3 62.8 62.3 61.9 65.1 59.4 ...
##  $ table  : num [1:53940] 55 61 65 58 58 57 57 55 61 61 ...
##  $ price  : int [1:53940] 326 326 327 334 335 336 336 337 337 338 ...
##  $ x      : num [1:53940] 3.95 3.89 4.05 4.2 4.34 3.94 3.95 4.07 3.87 4 ...
##  $ y      : num [1:53940] 3.98 3.84 4.07 4.23 4.35 3.96 3.98 4.11 3.78 4.05 ...
##  $ z      : num [1:53940] 2.43 2.31 2.31 2.63 2.75 2.48 2.47 2.53 2.49 2.39 ...
```

Question 1

Assumptions

1. The data is made up of basic random values drawn from the population
2. First, we test the hypothesis of normality of the dependent variable therefore the price of diamonds

```
##
##  Shapiro-Wilk normality test
##
## data:  diamonds$price[1:5000]
## W = 0.66175, p-value < 2.2e-16
```

Since the p-value $<$ 2.2e-16, implies that data distribution don't follows the normal distribution.

3. Next, we will test the hypothesis of equality of variances between the groups.

```
##
##  Bartlett test of homogeneity of variances
##
## data:  diamonds$price by diamonds$cut
## Bartlett's K-squared = 406.7, df = 4, p-value < 2.2e-16
```

Since the p-value$<$ 2.2e-16, implies that the variances don't to be equal. Hence not assuming equal variances.

ANOVA

Let m1, m2, m3, m4 and m5 respectively be the mean prices of diamonds which group are "Ideal", "Premium", "Good", "Very Good", and "Fair".

The hypotheses are:

- H0: m1=m2=m3=m4=m5
- Ha: At least one mean different

```
##
##  One-way analysis of means (not assuming equal variances)
##
## data:  price and cut
## F = 166.04, num df = 4.0, denom df = 9398.6, p-value < 2.2e-16
```

Conclusion With an alpha of 0.05, and a p-value<2.2e-16, we reject the null and stat that we have sufficient evidence to show that there was a effect of cut on diamond prices.

Question 2

The hypotheses are:

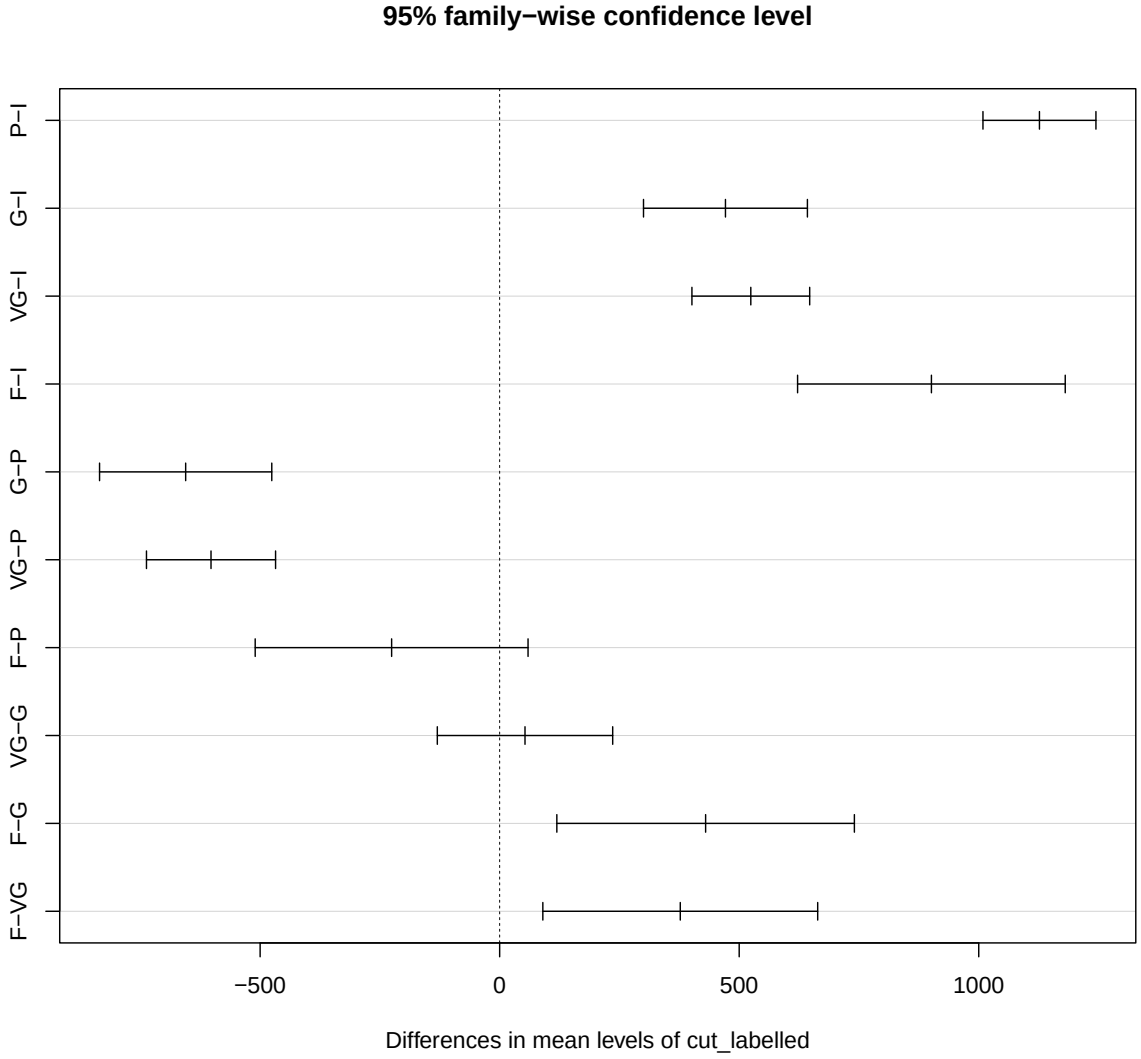
- H0: m1=m2=m3=m4=m5
- Ha: At least one mean different

```
##
##  One-way analysis of means
##
## data:  price and cut
## F = 175.69, num df = 4, denom df = 53935, p-value < 2.2e-16
```

There is no obvious difference in p-value but the F-statistic is different as well as "num df" and "denom df".

Question 3

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = price ~ cut_labelled, data = diamonds)
##
## $cut_labelled
##            diff       lwr        upr     p adj
## P-I  1126.71573 1008.8088 1244.62267 0.0000000
## G-I   471.32248  300.2823  642.36268 0.0000000
## VG-I  524.21792  401.3312  647.10467 0.0000000
## F-I   901.21579  621.8602 1180.57139 0.0000000
## G-P  -655.39325 -835.1353 -475.65120 0.0000000
## VG-P -602.49781 -737.2331 -467.76249 0.0000000
## F-P  -225.49994 -510.2665   59.26664 0.1950425
## VG-G   52.89544 -130.1519  235.94273 0.9341158
## F-G   429.89331  119.3378  740.44880 0.0014980
## F-VG  376.99787   90.1336  663.86215 0.0031094
```



Interpretations

Question 4

The hypotheses are:

- H0: m3=m4
- Ha: m3!=m4

```
##
##  One-way analysis of means
##
## data:  price and cut
## F = 0.65381, num df = 1, denom df = 16986, p-value = 0.4188
```

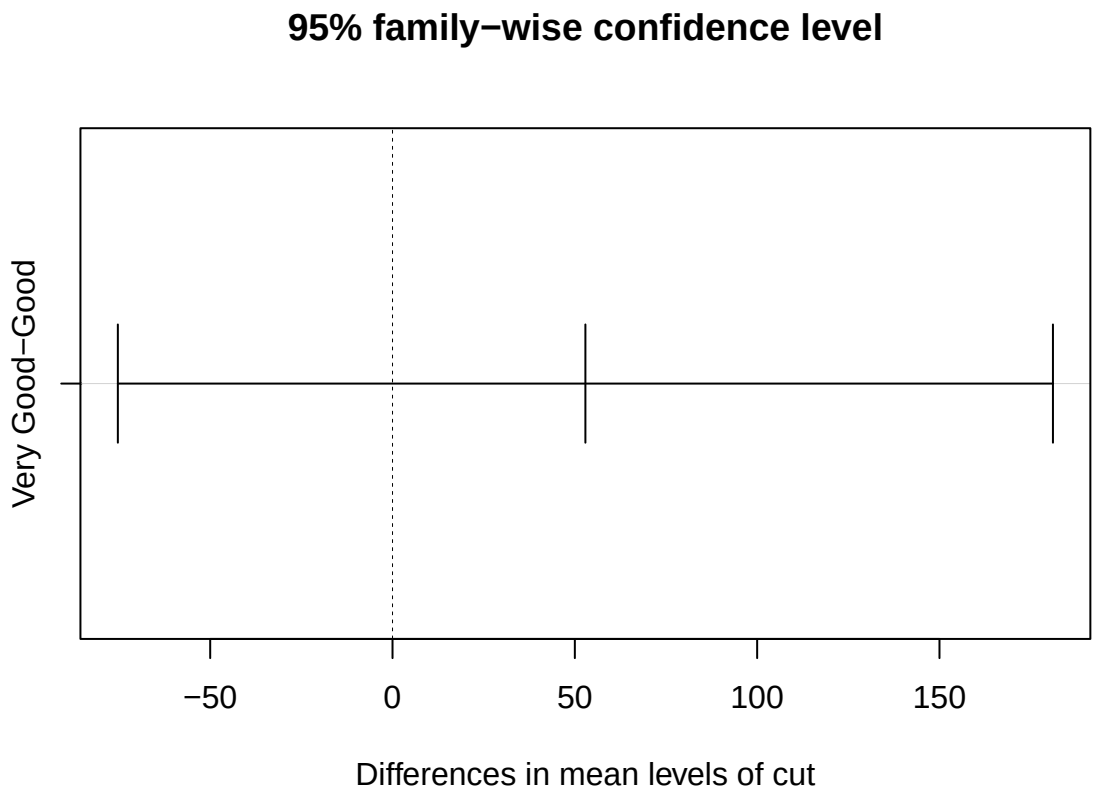
Conclusion With an alpha of 0.05, and a p-value of 0.4188, we fail to reject the null and stat that we have insufficient evidence to show that there was a cut effect of on diamond prices.

Question 5

```
model<-aov(price ~ cut, data = df)
TUKEY<-TukeyHSD(model, conf.level=0.95)
TUKEY
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = price ~ cut, data = df)
##
## $cut
##                     diff       lwr      upr     p adj
## Very Good-Good 52.89544 -75.32923 181.1201 0.4187652
```

```
plot(TUKEY)
```



Question 6

```
df <- subset(diamonds, select = c("cut","carat","color","depth","price"))
df$cut<- as.factor(df$cut)
df$color<- as.factor(df$color)
lr<-lm(price~cut+color+carat+depth, data = df)
summary(lr)
```

```
##
```

```
## Call:
## lm(formula = price ~ cut + color + carat + depth, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17265.2   -751.5    -83.6    544.2  12256.1
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -910.028    286.893  -3.172  0.00151 **
## cut.L        1177.948     26.101  45.130  < 2e-16 ***
## cut.Q        -491.180     22.520 -21.811  < 2e-16 ***
## cut.C         370.862     19.151  19.365  < 2e-16 ***
## cut^4          85.419     15.425   5.538 3.08e-08 ***
## color.L     -1572.449     21.727 -72.373  < 2e-16 ***
## color.Q      -732.336     19.849 -36.896  < 2e-16 ***
## color.C      -109.973     18.631  -5.903 3.60e-09 ***
## color^4        82.481     17.106   4.822 1.43e-06 ***
## color^5      -137.643     16.174  -8.510  < 2e-16 ***
## color^6      -161.317     14.672 -10.995  < 2e-16 ***
## carat        8183.774     13.889 589.216  < 2e-16 ***
## depth         -35.979      4.602  -7.819 5.43e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1431 on 53927 degrees of freedom
## Multiple R-squared:  0.8713, Adjusted R-squared:  0.8713
## F-statistic: 3.042e+04 on 12 and 53927 DF,  p-value: < 2.2e-16
```

Question 7

```
fitAll <- lm(price ~ ., data = subset(diamonds, select = -c(cut_labelled)))

lr.final <- step(fitAll, direction = "forward")
```

```
## Start:  AIC=758426.5
## price ~ carat + cut + color + clarity + depth + table + x + y +
##     z
```

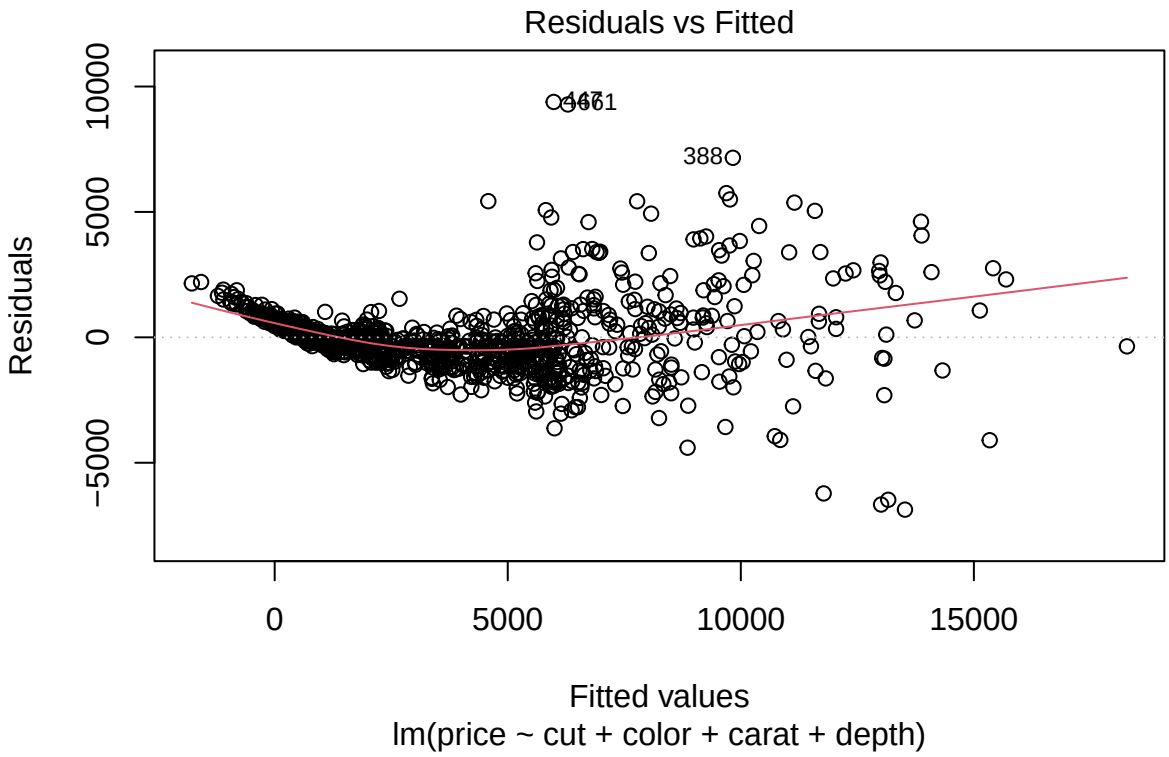
```
summary(lr.final)
```

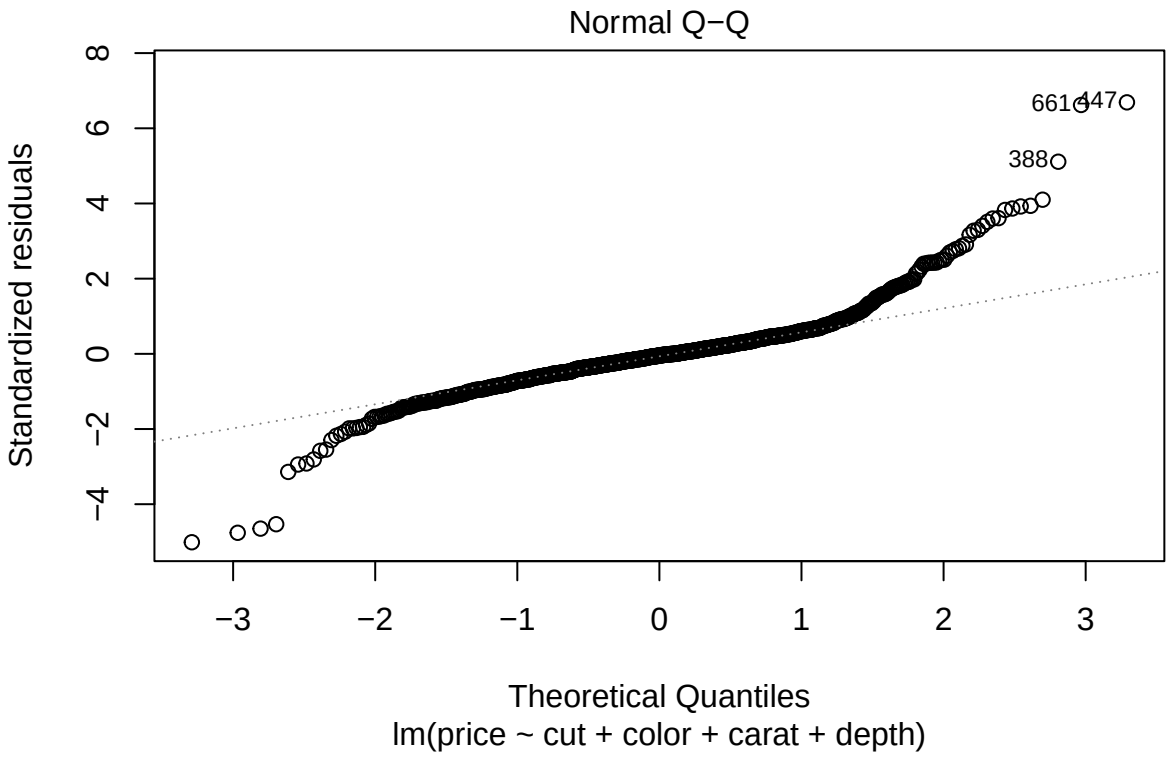
```
##
## Call:
## lm(formula = price ~ carat + cut + color + clarity + depth +
##     table + x + y + z, data = subset(diamonds, select = -c(cut_labelled)))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21376.0   -592.4   -183.5    376.4  10694.2
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept)  5753.762    396.630   14.507  < 2e-16 ***
## carat       11256.978     48.628  231.494  < 2e-16 ***
```

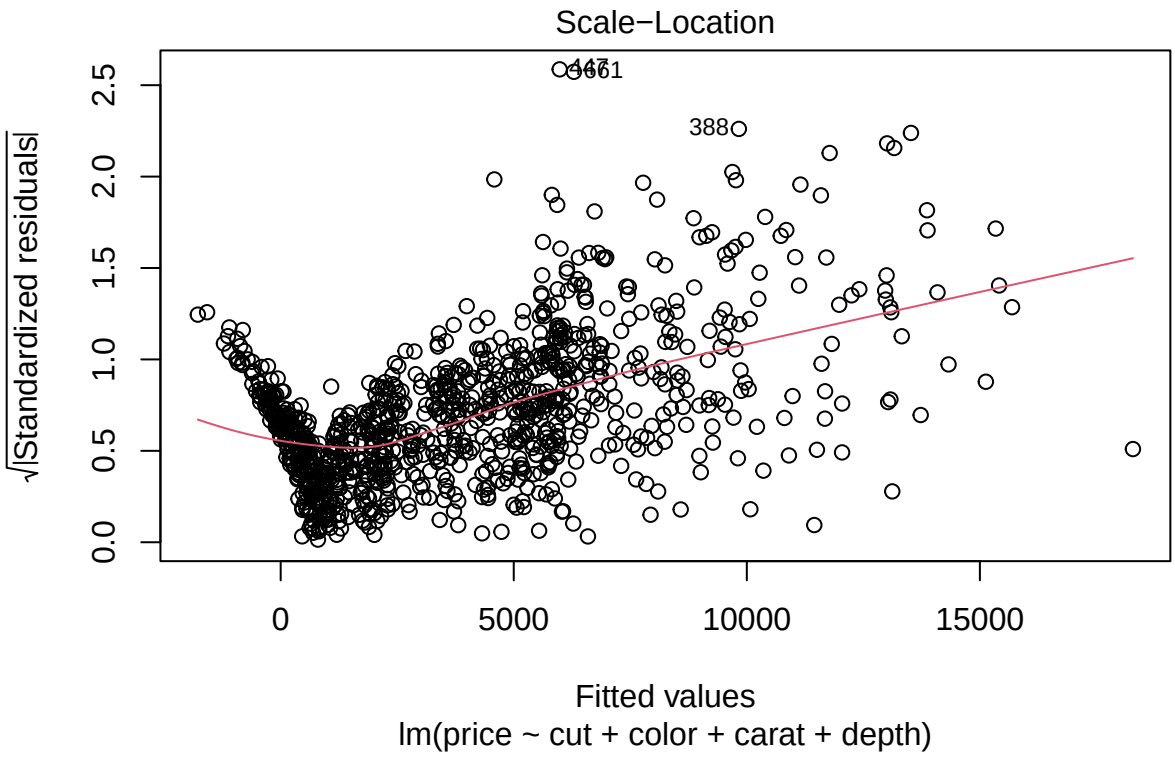
```
## cut.L         584.457     22.478   26.001  < 2e-16 ***
## cut.Q        -301.908     17.994  -16.778  < 2e-16 ***
## cut.C         148.035     15.483    9.561  < 2e-16 ***
## cut^4         -20.794     12.377   -1.680  0.09294 .
## color.L     -1952.160     17.342 -112.570  < 2e-16 ***
## color.Q      -672.054     15.777  -42.597  < 2e-16 ***
## color.C      -165.283     14.725  -11.225  < 2e-16 ***
## color^4        38.195     13.527    2.824  0.00475 **
## color^5       -95.793     12.776   -7.498 6.59e-14 ***
## color^6       -48.466     11.614   -4.173 3.01e-05 ***
## clarity.L    4097.431     30.259  135.414  < 2e-16 ***
## clarity.Q   -1925.004     28.227  -68.197  < 2e-16 ***
## clarity.C     982.205     24.152   40.668  < 2e-16 ***
## clarity^4    -364.918     19.285  -18.922  < 2e-16 ***
## clarity^5     233.563     15.752   14.828  < 2e-16 ***
## clarity^6       6.883     13.715    0.502  0.61575
## clarity^7      90.640     12.103    7.489 7.06e-14 ***
## depth         -63.806      4.535  -14.071  < 2e-16 ***
## table         -26.474      2.912   -9.092  < 2e-16 ***
## x           -1008.261     32.898  -30.648  < 2e-16 ***
## y               9.609     19.333    0.497  0.61918
## z             -50.119     33.486   -1.497  0.13448
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1130 on 53916 degrees of freedom
## Multiple R-squared:  0.9198, Adjusted R-squared:  0.9198
## F-statistic: 2.688e+04 on 23 and 53916 DF,  p-value: < 2.2e-16
```

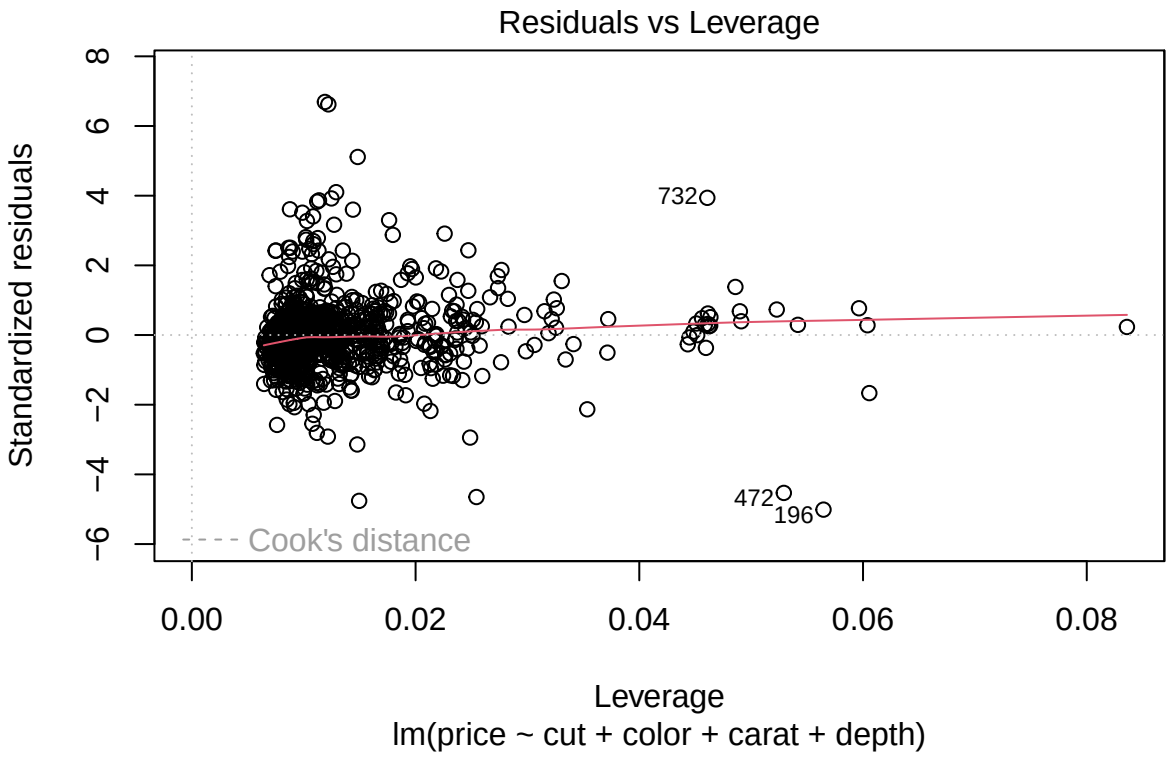
Question 8

```
df<-df[sample(nrow(df), 1000), ]
lr<-lm(price~cut+color+carat+depth, data = df)
plot(lr)
```


Residuals vs Fitted
Residuals
10000
5000
0
−5000
388
0
5000
10000
15000
Fitted values
lm(price ~ cut + color + carat + depth)


Normal Q−Q
Standardized residuals
8
6
4
2
0
−2
−4
661
447
388
−3
−2
−1
0
1
2
3
Theoretical Quantiles
lm(price ~ cut + color + carat + depth)


Scale−Location
√|Standardized residuals|
0.0
0.5
1.0
1.5
2.0
2.5
0
5000
10000
15000
Fitted values
lm(price ~ cut + color + carat + depth)


Residuals vs Leverage
Standardized residuals
8
6
4
2
0
−2
−4
−6
732
472
196
Cook's distance
0.00
0.02
0.04
0.06
0.08
Leverage
lm(price ~ cut + color + carat + depth)